

Question: 1 SQL Joins.

1.1. Inner Join.

Employee-ID	First-name	Last-name	Department	Salary	Project-ID	Project-name
1	John	Doe	IT	70000	101	AI - Devel
2	Alice	Smith	HR	60000	102	Emp - Train
3	Bob	Johnson	Finance	75000	104	Finan. Anal
5	Emma	Wilson	Sales	65000	105	Market - Exp
6	Michael	Clark	Finance	80000	106	Risk - Manag
1	David	Brown	IT	72000	103	Secur. Sys

Budget	Status
100000	Completed
50000	Ongoing
90000	Ongoing
65000	Completed
80000	Pending
Null	Null

1.2. Left Join.

Employee-ID	First-name	Last-name	Department	Salary	Project-ID	Project-name
1	John	Doe	IT	70000	101	AI - Devel
2	David	Brown	Null		Null	Null
3	Alice	Smith	HR	60000	102	Emp - Train
5	Bob	Johnson	Finance	75000	104	Finan. Dev
6	Emma	Wilson	Sales	65000	105	Market - Exp
6	Michael	Clark	Finance	80000	106	Risk Manag

Budget	Status
100000	Completed
50000	Ongoing
75000	Ongoing
65000	Completed
80000	Pending

1.3 Right Join.

Project-ID	Employee-ID	Project-name	Budget	Status	First Employee Name	Last-name	Department
101	1	AI - Devel	100000	Completed	John	Doe	IT
102	2	Emp training	50000	Ongoing	Alice	Smith	HR
103	3	Finan. Dev	90000	Ongoing	Bob	Johnson	Finance
104	5	Market - Exp	65000	Completed	Emma	Wilson	Sales
106	6	Risk manage	80000	Pending	Michael Clark	Clark	Finance
Null	Null	Secur. Sys	Null				

1.4.

Full Join

EmpID	First-name	Last-name	Department	Salary	Project-ID	Project-name
EmpID	First-name	Last-name	Department	Salary	Project-ID	Project-name
1	John	Doe	IT	70000	101	IT - Devel
2	Alice	Smith	HR	60000	102	Emp - Train
1	David	Johnson	Finance	75000	Null	WLC
3	Bob	Johnson	Finance	75000	103	Finan. Analy
5	Emma	Wilson	Sales	65000	105	market. Exp
6	Michael	Clark	Finance	80000	106	Risk - Manage
					Budget	Status
					100000	Completed
					80000	Ongoing
					Null	Null
					65000	Completed
					80000	Pending

3.

7.

Employee-ID	First-Name	Last-Name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

2.

Employee-ID	First-Name	Last-Name	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

11. Project - name Budget Status.

Employee - ID	First - name	Last - name	
103	Development	1	Pending
103	Cybersecurity	25000	ongang
104	Financial - analysis	90000	Pending
106	Risk management	80000	

12.

Employee - ID	First - name	Last - name	Department	Salary	City
1	John	Doe	IT	70000	New York
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12.

Employee - ID	First - name	Last - name	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
1	John	Doe	IT	70000

Aggregate Functions with Groupby & Having

13.

Department	Total - Salary
IT	R 142 000
Finance	R 156000
HR	R 60000
Sales	R 66000

14.

City	Average - Salary
Sydney	65000
New York	70000 - 80000
Toronto	75000

15.

Department	Employee Count
IT	2
HR	1
Finance	2

9. Retrieve projects that are **not yet completed**.

- Return: ProjectID, ProjectName, Budget, Status.

10. Retrieve projects that have a **budget greater than 70,000** and are **not completed**.

- Return: ProjectID, ProjectName, Budget, Status.

11. Retrieve employees from **New York OR Toronto**, ordered by **salary in descending order**.

- Return: EmployeeID, FirstName, LastName, Department, Salary, City.

12. Retrieve the **top 3 highest-paid employees**.

- Return: EmployeeID, FirstName, LastName, Department, Salary.

4. Aggregate Functions with GROUP BY & HAVING

13. Find the **total salary per department**, sorted in descending order.

- Return: Department, TotalSalary (Rename SUM(Salary) as TotalSalary).

14. Find the **average salary per city**, but only include cities where the average salary is greater than **65,000**.

- Return: City, AverageSalary (Rename AVG(Salary) as AverageSalary).

15. Count the **number of employees per department**, including only departments with more than 1 employee.

- Return: Department, EmployeeCount (Rename COUNT(EmployeeID) as EmployeeCount).

16. Retrieve the **number of projects per status**, but only include statuses with **at least 2 projects**.

- Return: Status, ProjectCount (Rename COUNT(ProjectID) as ProjectCount).

17. Retrieve the **total project budget per employee**, but only for employees who are managing projects worth more than **150,000**.

- Return: EmployeeID, FirstName, LastName, TotalProjectBudget (Rename SUM(Budget) as TotalProjectBudget).

a.

Project-ID	Project-name	Budget	Status
102	Employee-Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

SQL Practice Exercise: Joins, UNION, Filtering, and Aggregates

BrightLight Tutorials

Instructions:

1. Write answers on paper with a pen.
2. Draw tables of the final outcome.
3. In the SELECT statement, choose relevant columns to display, if not specified.
4. Scan the completed work into a PDF file and email it to:
rophiwa@brightlighttutorials.co.za
5. Submission Deadline: 25 March 2025, 00:00

Table 1: Employees

EmployeeID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Table 2: Projects

ProjectID	EmployeeID	ProjectName	Budget	StartDate	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

16.

Status	Project Count
Completed	2
Ongoing	2
Pending	2